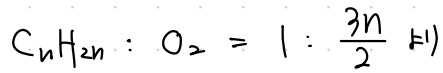
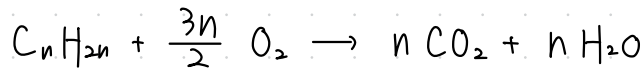


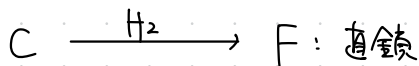
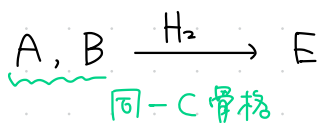
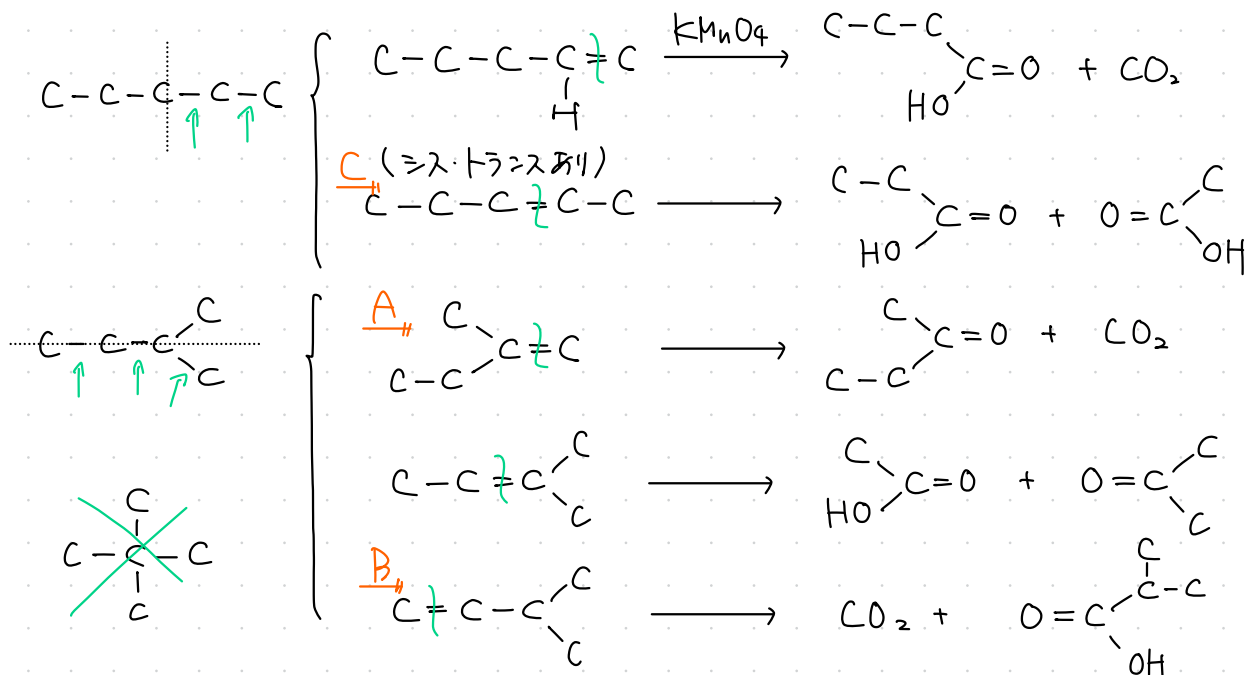
2-1

問1 (a), (b) は) A, B, C はアルケン  $C_nH_{2n}$

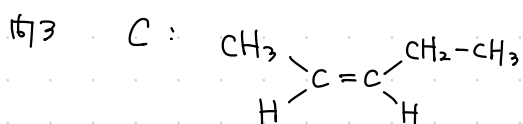
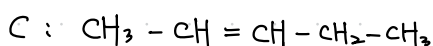
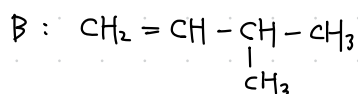
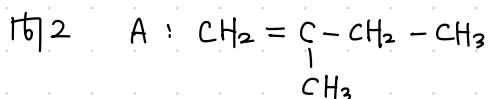


$$1 \text{ mol} \times \frac{3n}{2} = \frac{168 \text{ L}}{22.4 \text{ L/mol}} \therefore n = 5 \quad \underline{C_5H_{10}}$$

問5  $C_5H_{10}$   $U = 1$   $\therefore C=C \times 1$



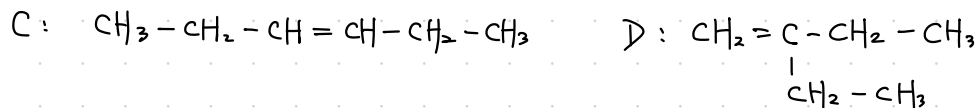
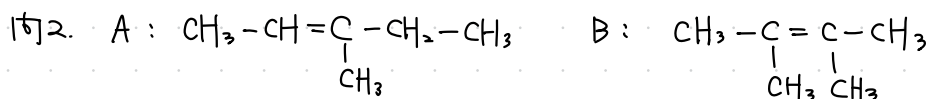
問1  $C_5H_{10}$



沸点  $\rightarrow$  分子間力  $\left\{ \begin{array}{l} \text{水素結合} \\ \text{ファンデルワールス力} \end{array} \right.$   
 $\Rightarrow$  ス型の方が極性  $\odot \rightarrow$  分子間力  $\odot$

2-2

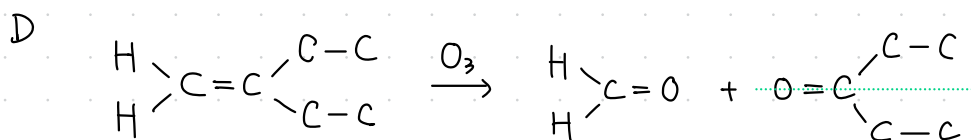
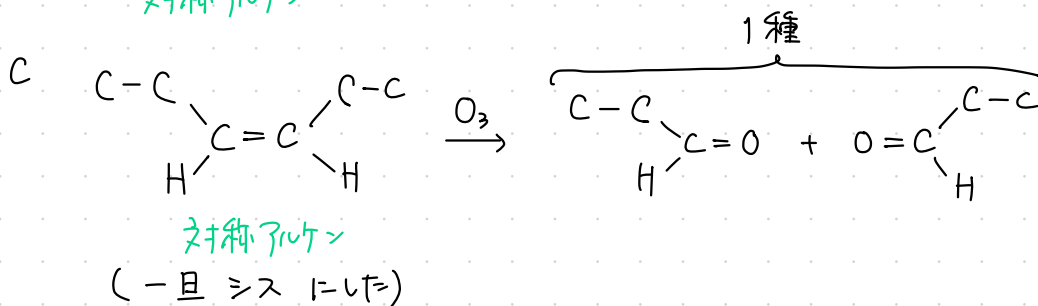
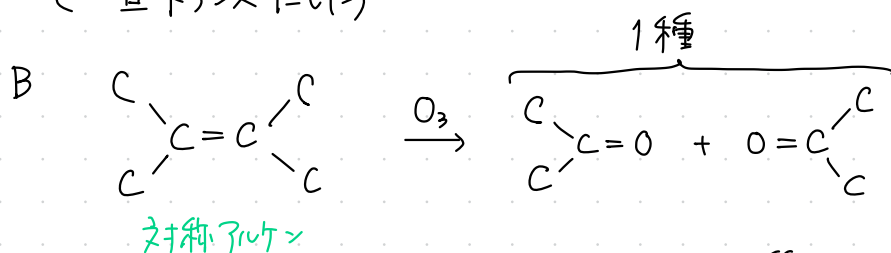
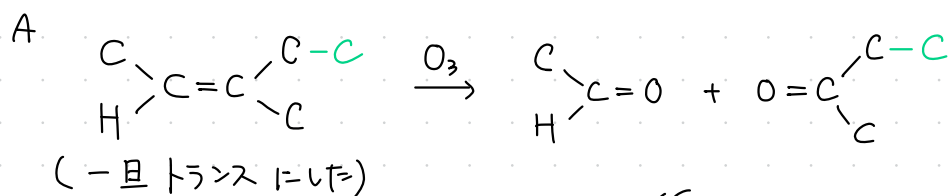
問1  $C_6H_{12}$



問3  $A \neq C$



$$\frac{2.1 \text{ g}}{14n \text{ g/mol}} = \frac{4.0 \text{ g}}{160 \text{ g/mol}} \quad \therefore n = 6 \quad \underline{C_6H_{12}}$$



2-3

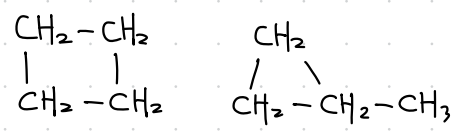
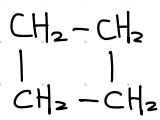
問1 A  $\text{CH}_2=\text{CH}-\text{CH}_2-\text{CH}_3$  1-ブテン

B  $\text{CH}_3-\text{CH}=\text{CH}-\text{CH}_3$  2-ブテン

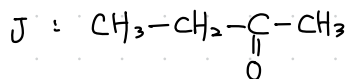
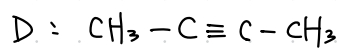
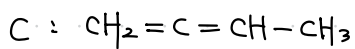
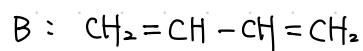
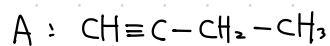
C  $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{CH}_3$  ブタン

問2 D: 2 E: 3 (メソ体あり)

問3

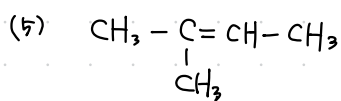
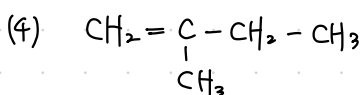
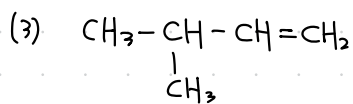


2 - 4



2 - 5

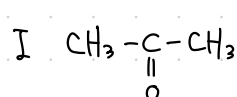
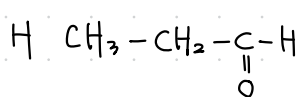
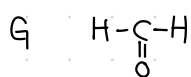
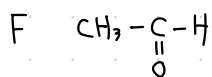
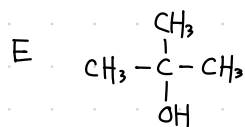
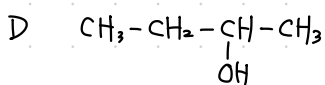
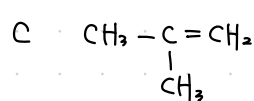
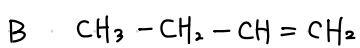
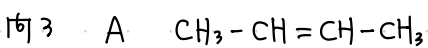
(1) 5 (2) 1



2 - 6

161 102

162  $\text{C}_4\text{H}_8$



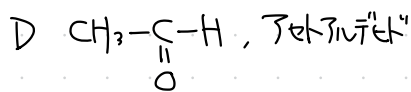
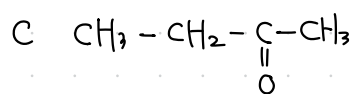
164 F

2 - [7]

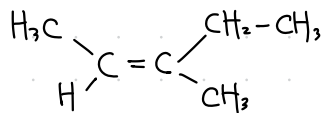
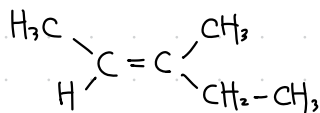
問1 略

問2 C  $C_4H_8O$  D  $C_2H_4O$

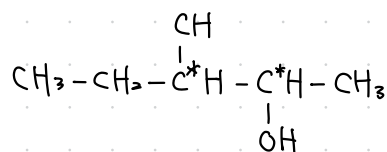
問3



問4.



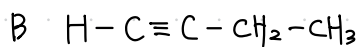
問5



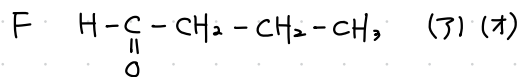
2 - [8]

問1 アセチレン,  $\gamma$ -ブチレン

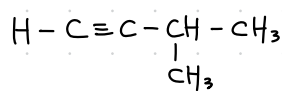
問2 (1) A  $CH_3-C \equiv C-CH_3$



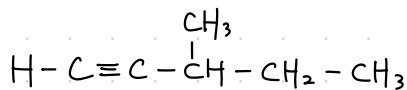
(2) E  $CH_3-\overset{\overset{O}{\parallel}}{C}-CH_2-CH_3$  (イ)



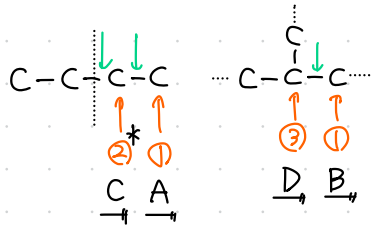
問3



問4.

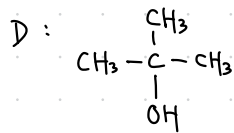
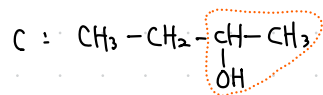
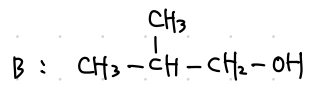
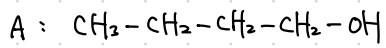


3-11  $C_4H_{10}O$   $U=1$   $\left\{ \begin{array}{l} \text{アルコール} \\ \text{エーテル} \end{array} \right.$



アルコール  $A^{\oplus} B^{\oplus} C^* D^{\ominus}$   
 $O_x$   $\underline{E} \quad \underline{F} \quad G \quad X$   
 $\text{アルコール} \quad \text{アルコール}$   
 b.p.  $A > B$

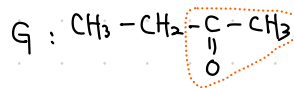
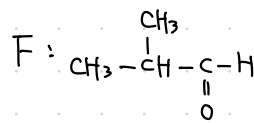
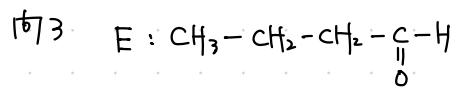
問1



問2

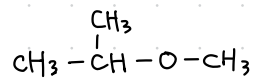
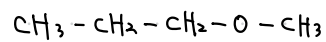
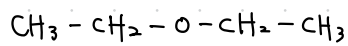
ヒドロキシ基を持ち、  
分子間での水素結合を形成するため。

問3

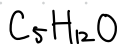


C, G

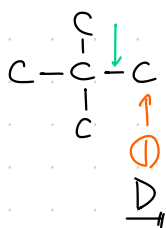
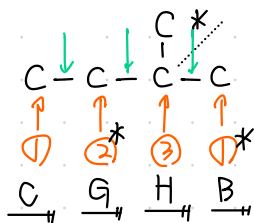
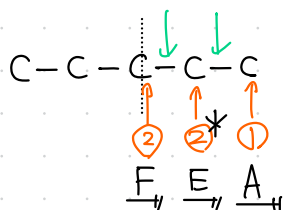
問4



3 - 2

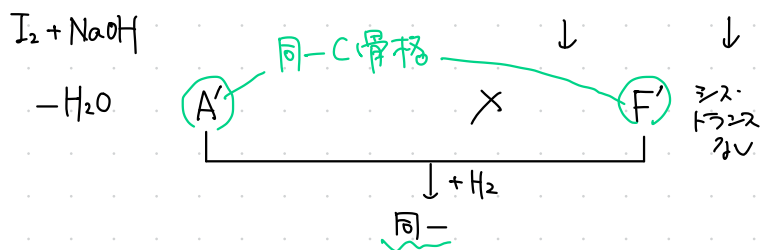


$$U=0 \quad \left\{ \begin{array}{l} \gamma_{\mu\nu} = -\gamma_{\nu\mu} \\ I = -\bar{I} \end{array} \right.$$



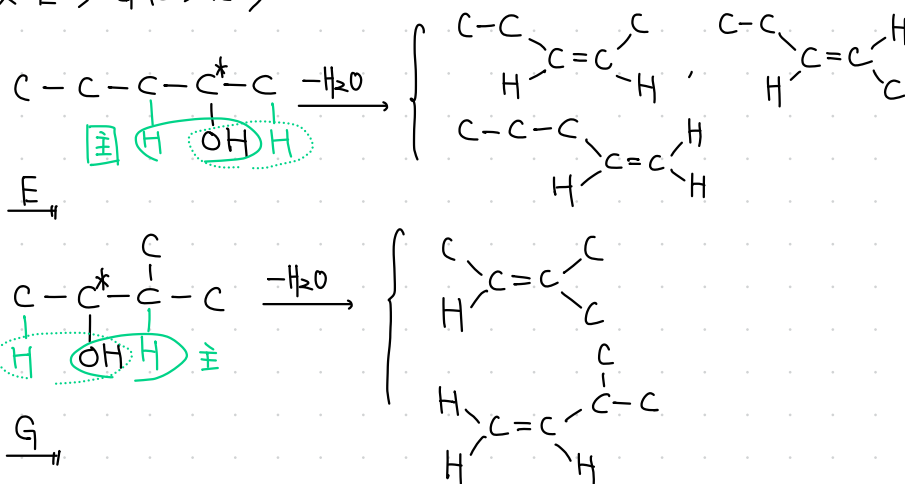
アルコール      ① A      ① B<sup>\*</sup>      ① C      ① D      ② E<sup>\*</sup>      ② F      ③ G<sup>\*</sup>      ③ H

Ox                  { アルデヒド }                  { ケトン }                  X



$\left. \begin{array}{l} \text{「脱水した} \sim \text{」} \\ \text{「H}_2\text{付加した} \sim \text{」} \end{array} \right\} \text{同一化合物}$   
 $\longrightarrow \text{同一C骨格}$

$\langle 2 \text{級 } E^*, G^* | \Rightarrow WZ \rangle$

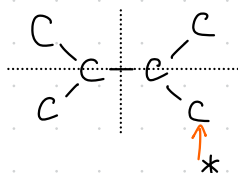
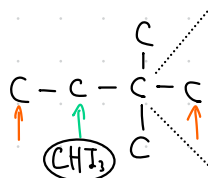
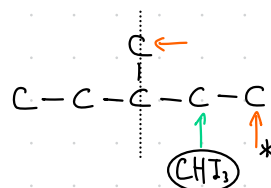
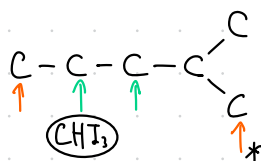
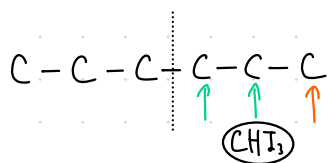


(2)  $\xrightarrow{6}$  (立体异构: 7)

3 - ③

← : =O (アルデヒド)  
← : =O (ケト>)

問1

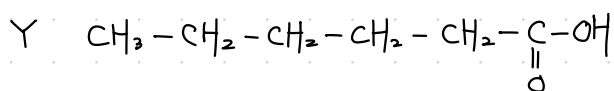
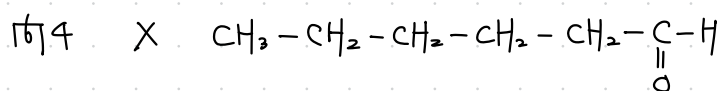
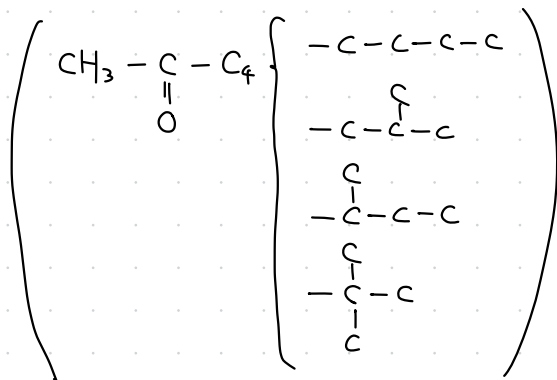


(a) 8

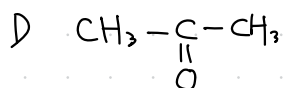
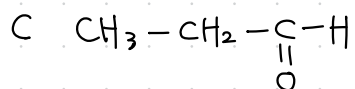
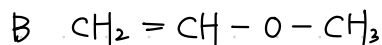
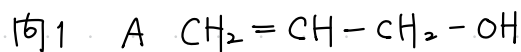
(b) 6

問2. 3

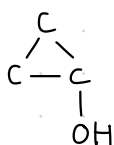
問3. 4



3 - ④

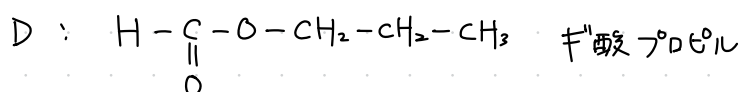
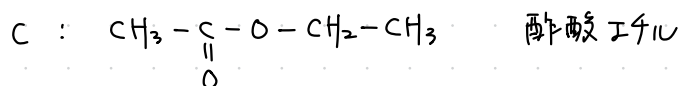
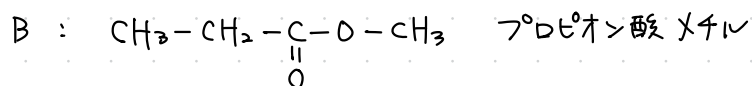
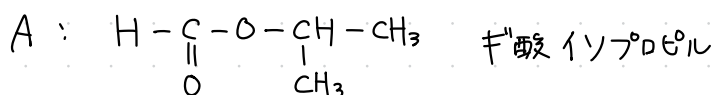
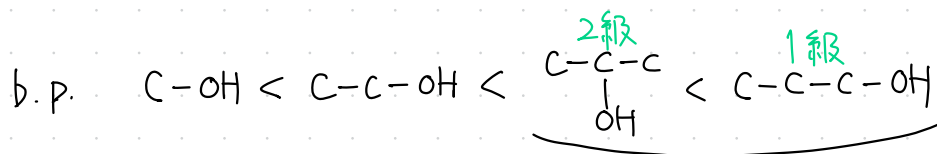
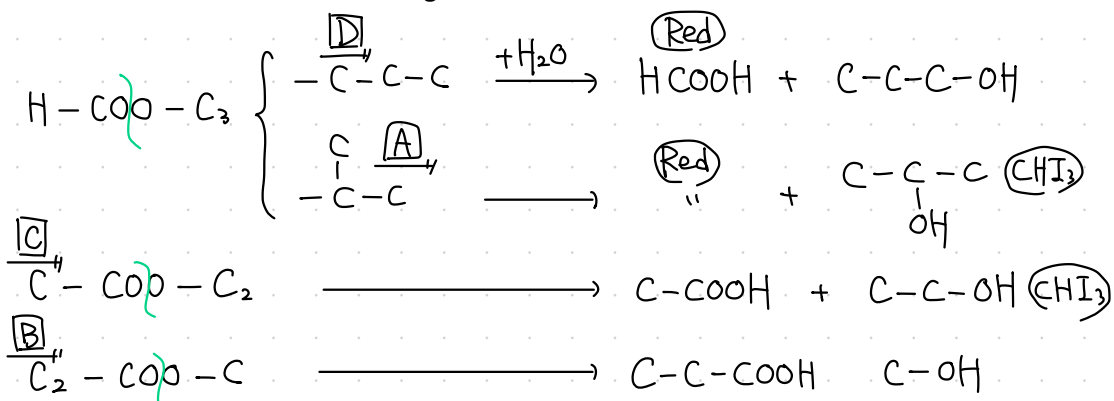
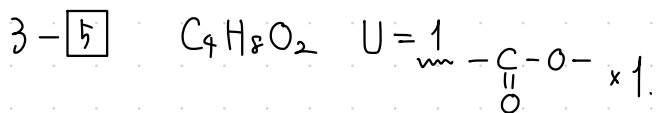


問2



3





3-6

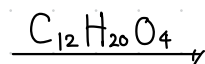
$$\text{例1} \quad w_C = 264 \text{ g} \times \frac{C(12)}{CO_2(44)} = 72 \text{ g}$$

$$w_H = 90 \text{ g} \times \frac{2H(2)}{H_2O(18)} = 10 \text{ g}$$

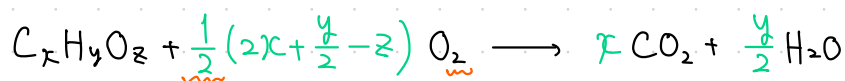
$$w_O = 114 - (72 + 10) = 32 \text{ g}$$

$$C : H : O = \frac{72 \text{ g}}{12 \text{ g/mol}} : \frac{10 \text{ g}}{1.0 \text{ g/mol}} : \frac{32 \text{ g}}{16 \text{ g/mol}} = 3 : 5 : 1$$

$$(C_3H_5O)_n = 57n = 228 \quad n = 4$$



<別解> 燃烧例, 分子量既知の時



$$\begin{matrix} C_xH_yO_z : CO_2 = 1 : x \text{ mol} \\ (288) \quad \quad (44) \end{matrix}$$

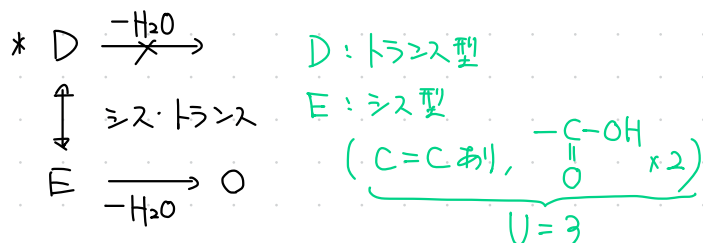
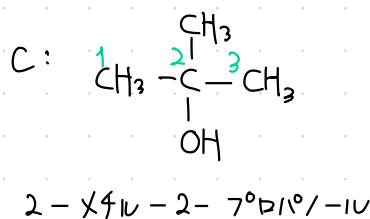
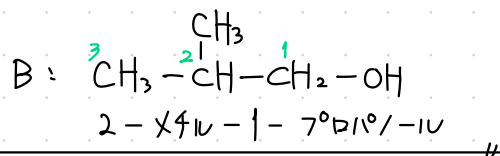
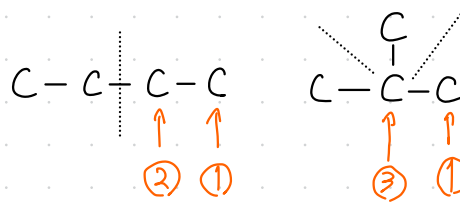
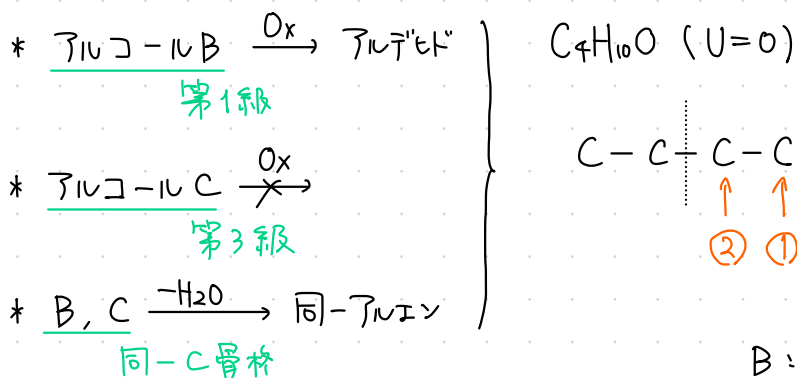
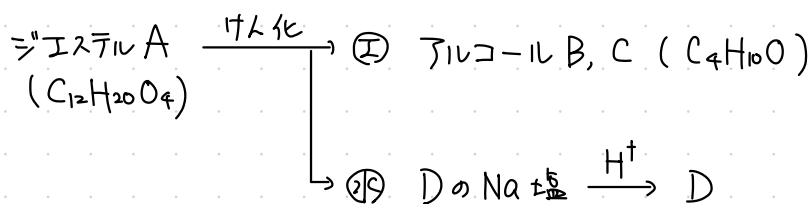
$$\frac{114 \text{ g}}{288 \text{ g/mol}} \times x = \frac{264 \text{ g}}{44 \text{ g/mol}} \times 1 \quad \therefore x = 12$$

$$\begin{matrix} C_xH_yO_z : H_2O = 1 : \frac{y}{2} \text{ mol} \\ (288) \end{matrix}$$

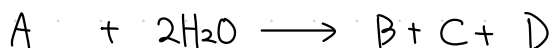
$$\frac{114 \text{ g}}{288 \text{ g/mol}} \times \frac{y}{2} = \frac{90 \text{ g}}{18 \text{ g/mol}} \times 1 \quad \therefore y = 20$$

$$C_{12}H_{20}O_z = 164 + 16z = 228 \quad \therefore z = 4 \quad \underline{C_{12}H_{20}O_4}$$

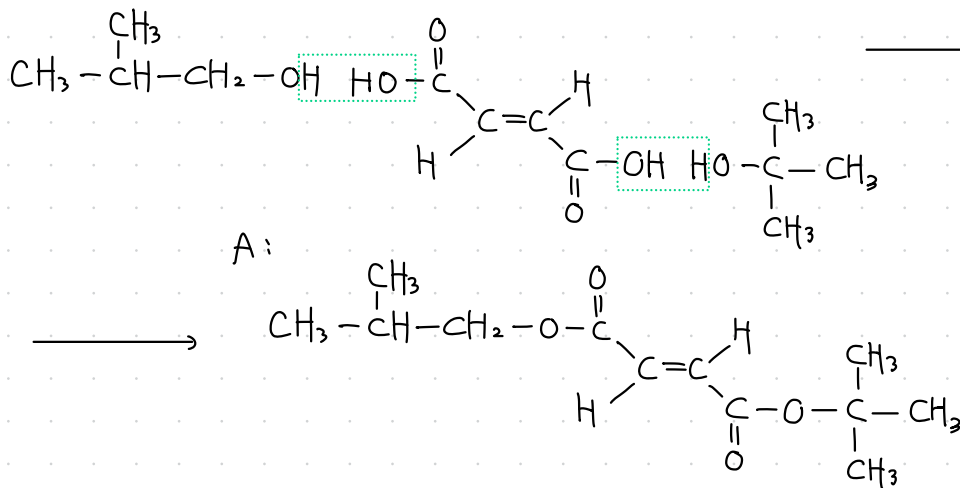
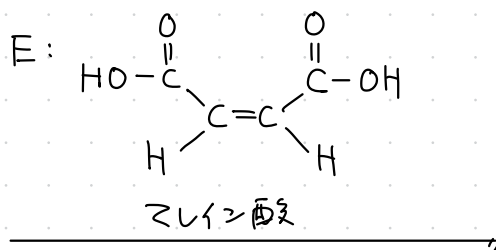
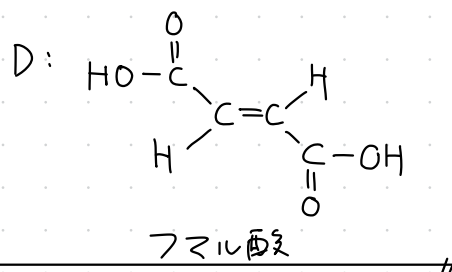
問2. A:  $C_{12}H_{20}O_4$   $U = \frac{2 \times 12 + 2 - 20}{2} = 3$  C=C  $\times 1$  +  $-\overset{\text{O}}{\underset{\text{O}}{\text{C}}}-\text{O}- \times 2$



ジエステル



$\therefore D = A + 2H_2O - (B + C)$   
 $= C_{12}H_{20}O_4 + 2H_2O - 2 \times C_4H_{10}O$   
 $= C_4H_4O_4$  (114)  
 ( $U=3$ )



3 - [7]

問1  $C_7H_8O_4$

$$C:H:O = \frac{53.8}{12} : \frac{5.1}{1.0} : \frac{41.1}{16} = 4.48 : 5.1 : \frac{2.56}{\text{①}}$$

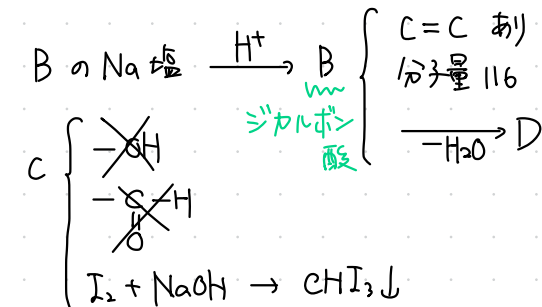
$$\div 1.75 : 1.99 : 1 = 7:8:4$$

$$130 \leq (C_7H_8O_4)_n \leq 170 \quad \therefore n=1 \quad \underline{C_7H_8O_4} \quad (U=4)$$

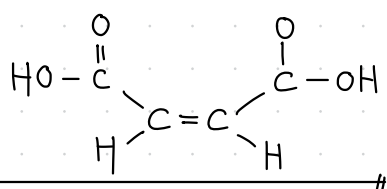
問2

酸性エステル  
A

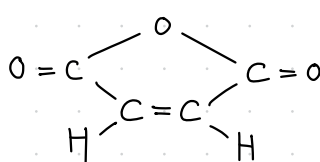
けん化



B:



D:



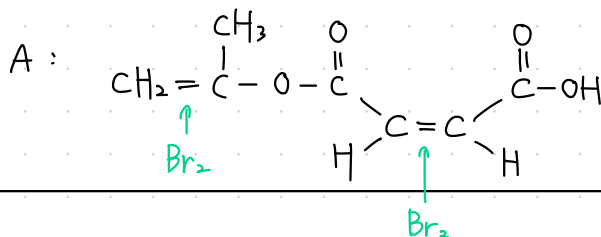
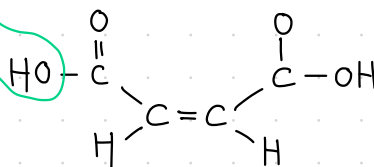
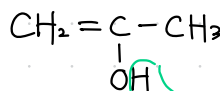
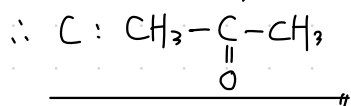
$$A + H_2O \longrightarrow B + C \quad \therefore C = A + H_2O - B$$

$$= C_7H_8O_4 + H_2O - C_4H_4O_4$$

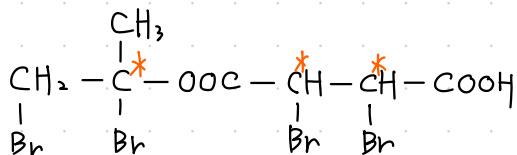
$$= \underline{C_3H_6O} \quad (U=1)$$

エステルの加水分解生酸性物として  $-OH$  基を持つものがない  
→ エノール型が ケト型 に変わった。

(ケト型)

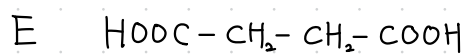
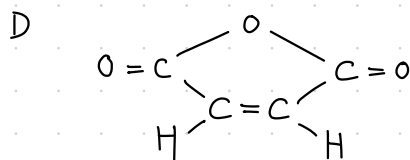
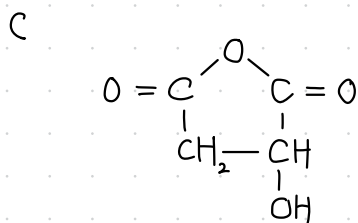
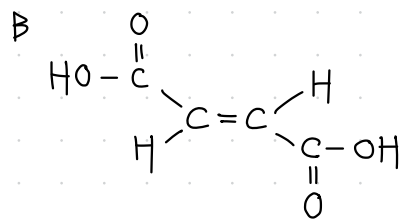
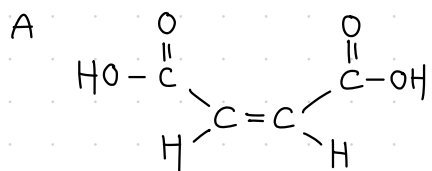


問3



$$2^3 = \underline{8}$$

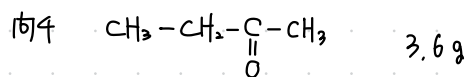
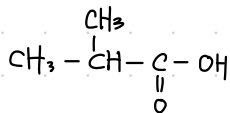
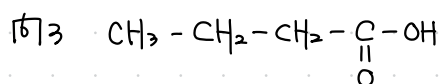
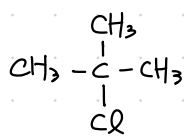
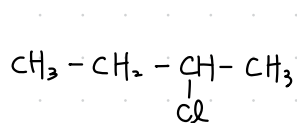
3-8



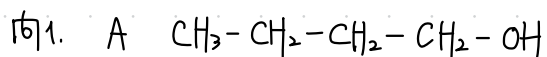
3-9

1 4

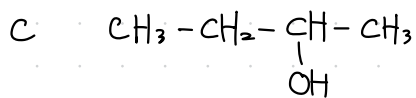
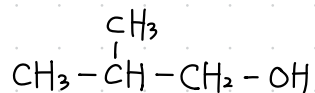
2.



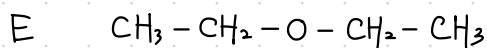
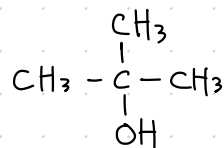
3-10



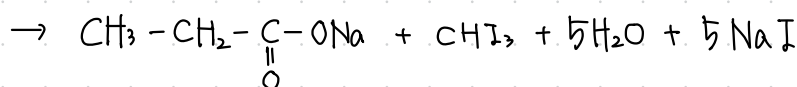
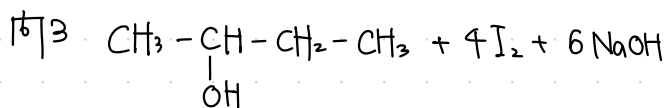
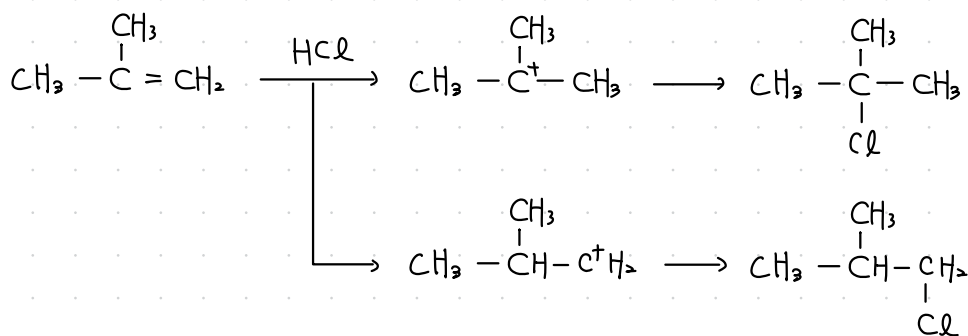
B



D



問2

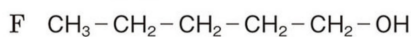
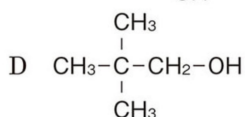
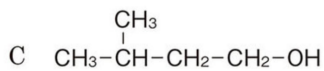
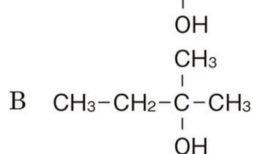


問4 C

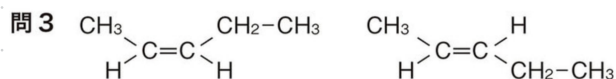
問5 エーテル系結合

3-11

問1A  $\text{CH}_3-\text{CH}_2-\underset{\text{OH}}{\underset{|}{\text{CH}}}-\text{CH}_2-\text{CH}_3$



問2 K



問1 A  $\text{CH}_3\text{--CH}_2\text{--CH}_2\text{--CH}_2\text{--CH}_2\text{--OH}$ B  $\text{CH}_3\text{--CH}_2\text{--CH}_2\text{--}\overset{\text{OH}}{\underset{|}{\text{CH}}}\text{--CH}_3$ C  $\text{CH}_3\text{--CH}_2\text{--}\overset{\text{OH}}{\underset{|}{\text{CH}}}\text{--CH}_2\text{--CH}_3$  D  $\text{CH}_3\text{--CH}_2\text{--}\overset{\text{CH}_3}{\underset{\text{OH}}{\underset{|}{\text{CH}}}}\text{--CH}_2$ E  $\text{CH}_3\text{--CH}_2\text{--}\overset{\text{CH}_3}{\underset{\text{OH}}{\underset{|}{\text{C}}}}\text{--CH}_3$  F  $\text{CH}_3\text{--}\overset{\text{CH}_3}{\underset{\text{OH}}{\underset{|}{\text{CH}}}}\text{--CH--CH}_3$ 

問2 0.15 L 問3 銀鏡反応, アルデヒド基

問4  $\text{CH}_3\text{--CH}_2\text{--}\overset{\text{H}}{\underset{\text{H}}{\text{C}}}=\overset{\text{CH}_3}{\underset{\text{H}}{\text{C}}}$  ,  $\text{CH}_3\text{--CH}_2\text{--}\overset{\text{H}}{\underset{\text{H}}{\text{C}}}=\overset{\text{H}}{\underset{\text{CH}_3}{\text{C}}}$  $\text{CH}_3\text{--CH}_2\text{--CH}_2\text{--}\overset{\text{H}}{\underset{\text{H}}{\text{C}}}=\overset{\text{H}}{\underset{\text{H}}{\text{C}}}$ 

問5

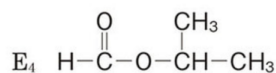
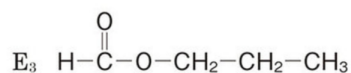
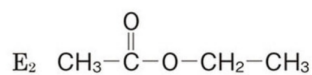
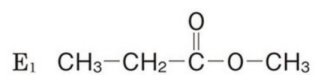
 $\text{H}_2\text{C--CH}_2\text{--CH}_2\text{--CH}_2\text{--CH}_2\text{--CH}_2$  $\text{H}_2\text{C--}\overset{\text{CH}_3}{\underset{\text{H}_2\text{C--CH}_2}{\text{CH}}}$  $\text{H}_3\text{C--}\overset{\text{CH}_3}{\underset{\text{H}_2\text{C--CH}_2}{\text{C}}}$  $\text{H}_2\text{C--}\overset{\text{CH--CH}_2\text{--CH}_3}{\text{CH}_2}$  $\text{H}_2\text{C--}\overset{\text{CH--CH}_3}{\text{CH--CH}_3}$ 

問6

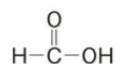
 $\text{CH}_3\text{--CH}_2\text{--}\overset{\text{CH}_3}{\underset{\text{OH Cl}}{\underset{|}{\text{C}}}}\text{--CH}_2$  $\text{CH}_3\text{--}\overset{\text{CH}_3}{\underset{\text{Cl}}{\underset{|}{\text{CH}}}}\text{--}\overset{\text{CH}_3}{\underset{\text{OH}}{\underset{|}{\text{C}}}}\text{--CH}_3$

3-13

問1



問2 ギ酸は分子内にホルミル基を持つため。(19字)

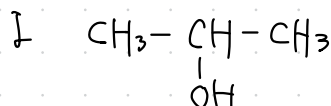
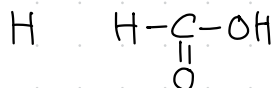
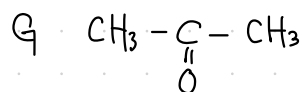
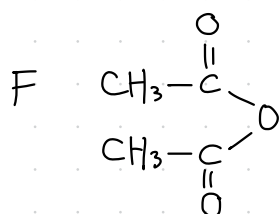
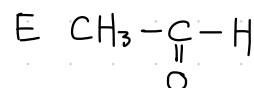
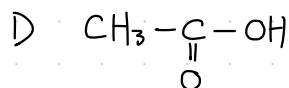
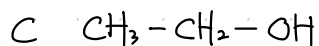
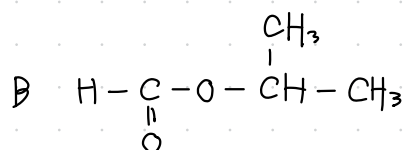
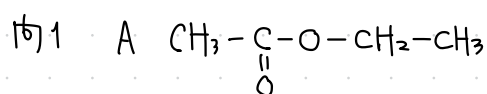


問3 酢酸カルシウム



問4 4.4 g

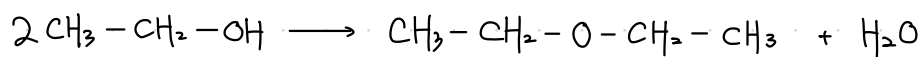
3-14



問2,3 略

問4. テトラアルデヒド

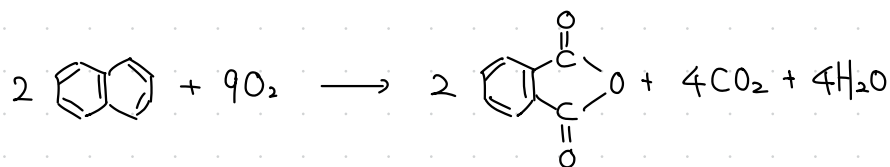
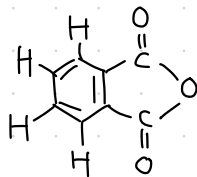
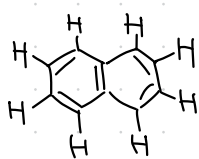
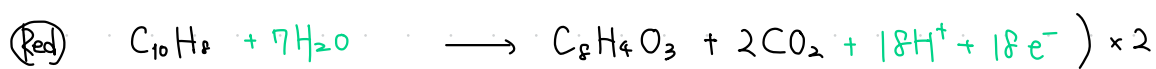
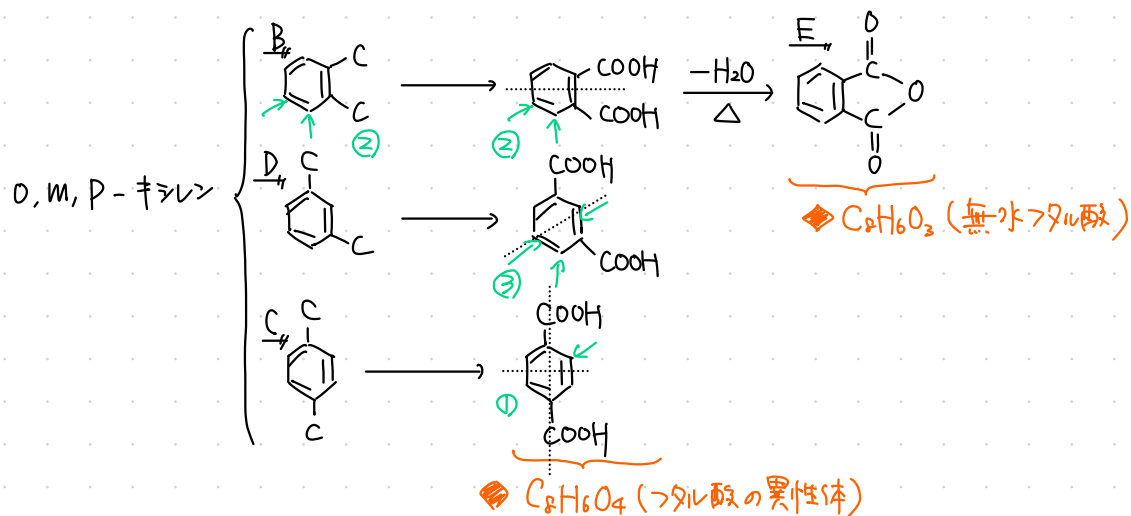
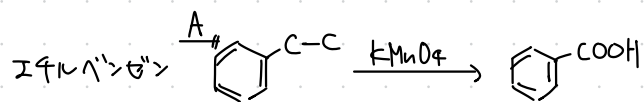
問5 シエチルエーテル







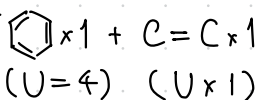
4-①  $C_8H_{10}$   $U = 4$    $\times 1$



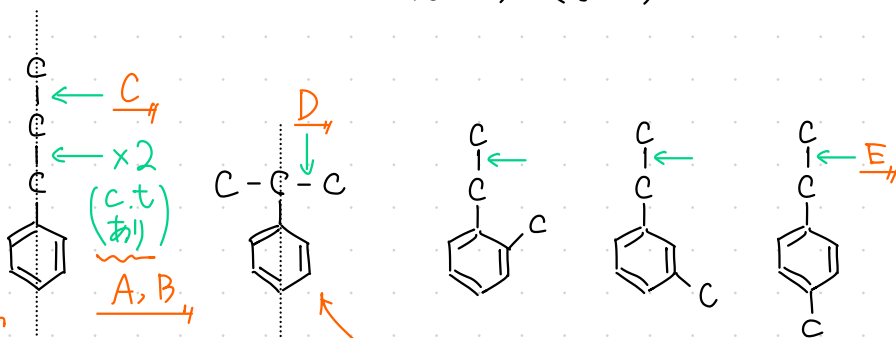
4-2

$C_9H_{10}$

$U = 5$

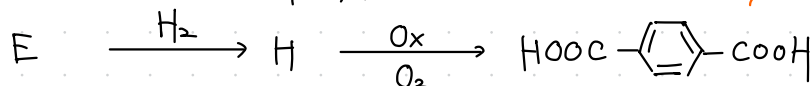
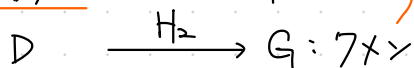


内2.



< 1置換体 >

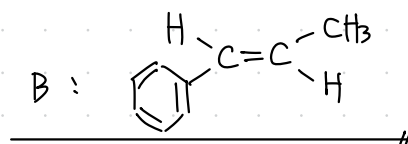
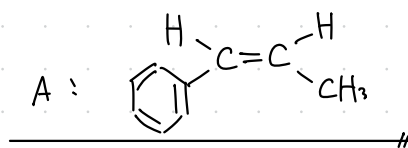
< 2置換体 >



A, B: シス・トランス異性体

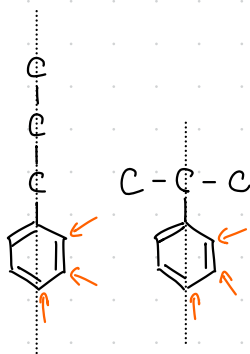
融点  $A < B$

シス トランス



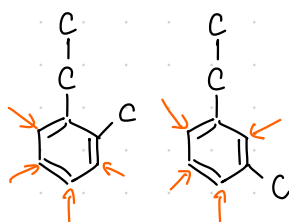
内1. 8

内3. 22



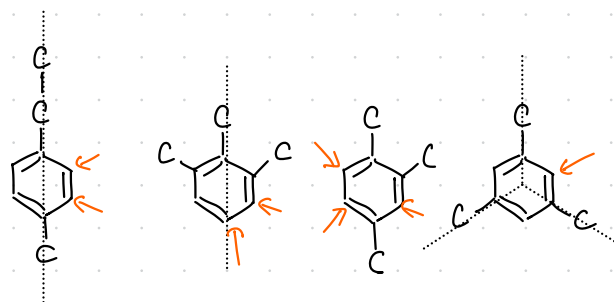
< 1置換体 >

Cl置換: 6



< 2置換体 >

Cl置換: 10



< 3置換体 >

Cl置換: 6

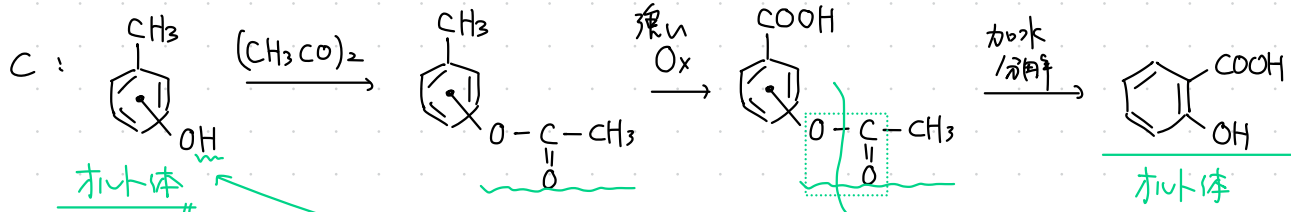
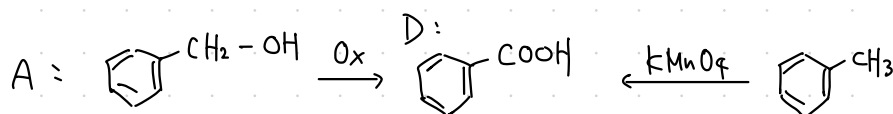
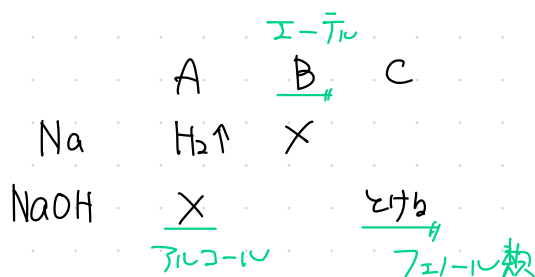
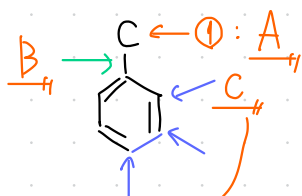
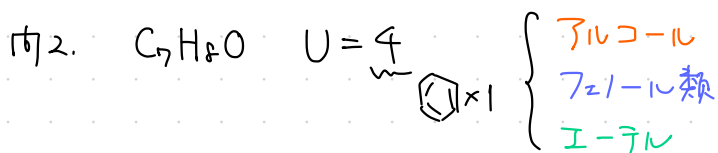
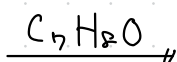
4-③

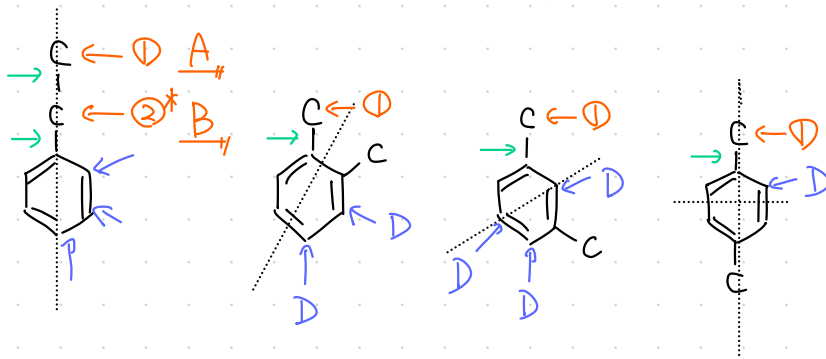
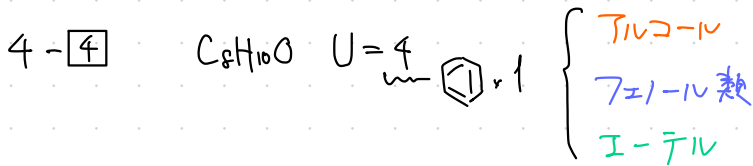


条件より  $l : \frac{m}{2} = 7 : 4 \quad \therefore (l, m) = (7, 8)$

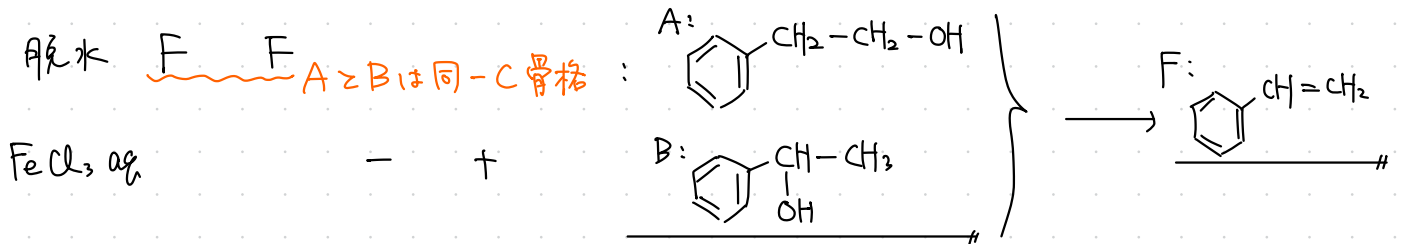
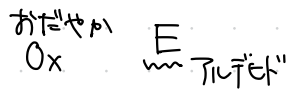
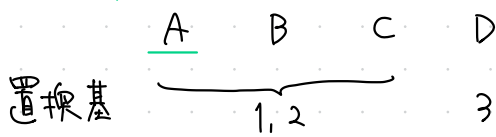
$\therefore (C_7H_8O_n)_k = (92 + 16n)k = 108$

$\therefore (n, k) = (1, 1)$

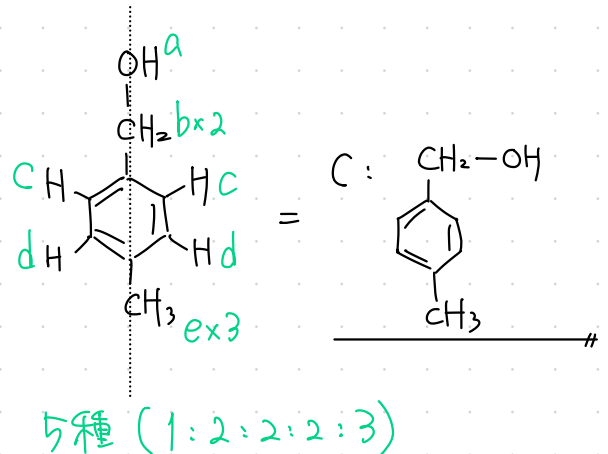
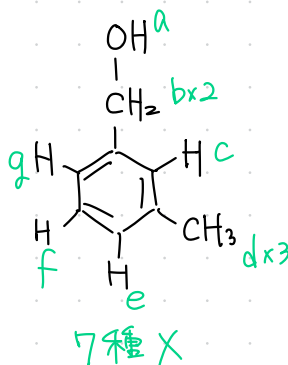
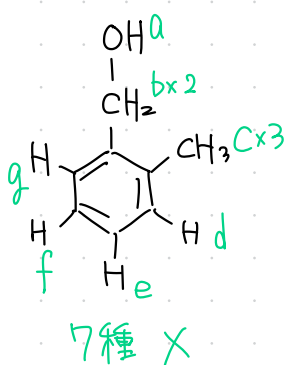




第1級  
アルコール



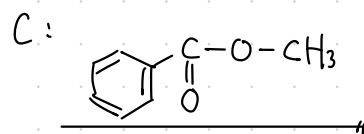
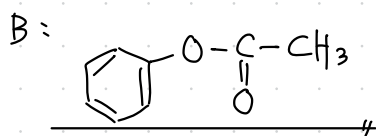
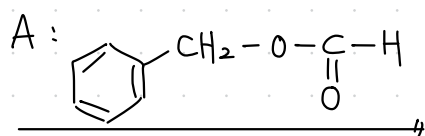
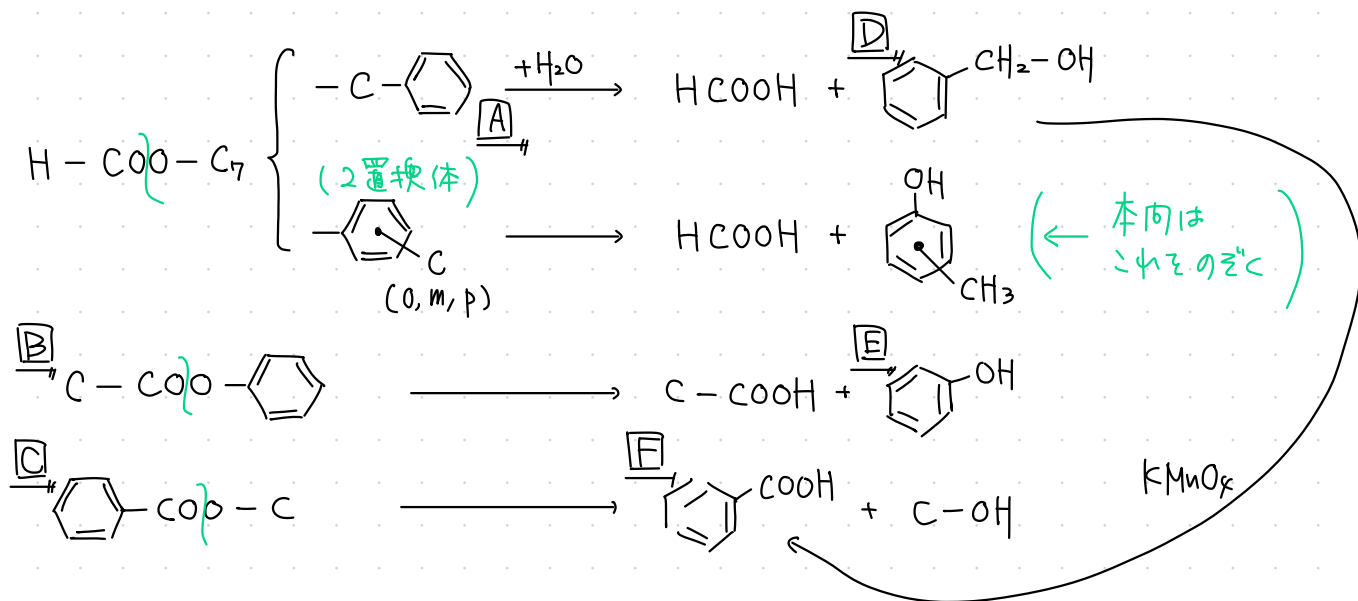
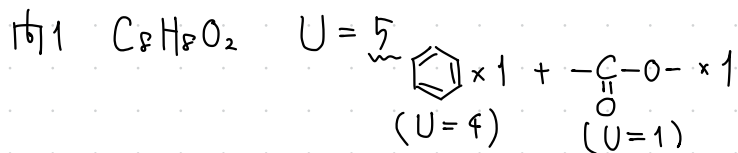
C : 1 or 2 置換体のアルコール



内2

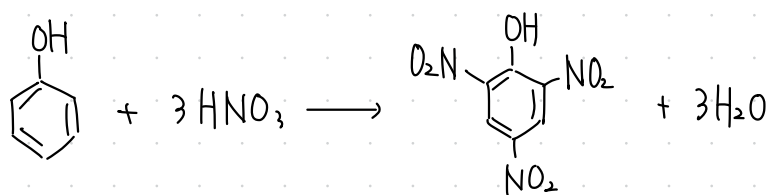
D: フェニル類 (3 置換体),  $\underline{6}$

4-7



162. E

163



ヒョウリ=西

2, 4, 6 - トニトロフェノール